

SACHI KAUSHIK

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EDUCATION

Columbia University

M.S. in Computer Science, Machine Learning Track, GPA: 3.52/4.0

New York, NY

Exp Dec 2025

Vellore Institute of Technology

B.Tech. in Computer Science and Engineering, GPA: 9.32/10.0

Vellore, IN

May 2024

WORK EXPERIENCE

adMarketplace

New York, NY

Machine Learning Engineer Intern

Jun 2025 – Present

- Training transformer-based embedding models and integrating FAISS similarity search on 10M+ vectors with real-time streaming pipelines to enable scalable CTR prediction and optimization.
- Improved a time series CTR prediction model by incorporating demographic features like age, achieving a 24% reduction in RMSE and demonstrating measurable gains in personalization effectiveness.
- Modeled DistilGPT-2 on partial-to-complete query term pairs for real-world query expansion, increasing correct expansions by 17% and enabling faster, more relevant ad matches from incomplete inputs.

Columbia University

New York, NY

Research Assistant

Jan 2025 – May 2025

- Architected a scalable ML pipeline for YouTube sentiment analysis (10k comments, 500 videos) using automated scraping, transcript parsing, and supervised sentiment modeling, reaching ~88% accuracy.
- Facilitated stance classification and OpenAI-based summarization, generating structured insights that surfaced 15% higher negative sentiment than baseline analysis, informing data-driven policy evaluation.

Tech Mahindra Maker's Lab

Pune, IN

Software Development Intern

May 2023 – Aug 2023

- Designed a production-grade Knowledge Management portal chatbot with scalable API architecture and low-latency response routing, improving information access workflows and reducing response times by 40%.
- Integrated a next-word recommendation system (100k+ users) trained on internal query logs using language modeling and LSTM sequence prediction, reducing average typing length per query by 12%.

TECHNICAL SKILLS

Coursework: Applied Machine Learning, Advanced Database Systems, Computer Vision, LLM-based Generative AI Systems

Programming Languages: C, C++, Python, R, Java, Matlab, HTML5, CSS, JavaScript, SQL, LaTeX

Libraries & Frameworks: OpenCV, scikit-learn, pandas, matplotlib, ONNX, PyTorch, TensorFlow, Flask, YOLOv7

Software & Tools: MySQL, Databricks, Azure, Git, GCP, AWS, Node.js, Oracle, Docker, Linux, Spark, CI/CD

PROJECTS

Intelligent Agentic RAG for Health Policy

Aug 2025 – Dec 2025

- Built a LangGraph multi-agent workflow (Classifier, Section-Analyzer, Entity-Extractor, Table-Extractor, Validator) to automate analysis of 100+ NYS health policy documents, reducing manual review by 85%.
- Improved retrieval and answer quality using BGE-M3 dense vectors, BM25 lexical search, OCR-based parsing, section-aware chunking, and encoder reranking, boosting policy Q&A accuracy by ~20%.

Mining the Web: Iterative Information Extraction with Few-Shot Learning

Jan 2025 – May 2025

- Developed a Python-based web information extraction system using Iterative Set Expansion, integrating spaCy for entity recognition and Google Custom Search to iteratively process 50+ pages/run to expand relation tuples.
- Formulated dual extraction pipelines with SpanBERT and Google Gemini API (LLM), designing prompts and applying entity-type filtering to minimize API calls, achieving over 90% precision on top-k tuples.

Cloud Casino – Cloud-Based Microservice Casino Platform

Sep 2024 – Dec 2024

- Collaborated on a scalable casino web application built with AWS and FastAPI, featuring Google OAuth login, game stat tracking, and real-time analytics with < 200ms API latency across multiple microservices.
- Deployed 5 microservices across AWS EC2 and Google PaaS, configured API Gateway with JWT authentication blocking 100% unauthorized requests, and automated signup notifications via SES + Lambda.